Project Plan

Smart Mini Terrarium 🌱

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Sisällys

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1.1 Topic 2
1.1.1 Wireframes ..... **Error! Bookmark not defined.**

Project Plan

The project automatically keeps humidity within a target range AND support remote dashboard control.

1.1 Topic

The Smart Mini Terrarium consists of:

- A **Raspberry Pi RP2040 (Pico W)** that collects sensor data and controls hardware
- A **Qt application** (MCU) that communicates with the RP2040 using Wi-Fi
- A humidity sensor, mist maker, and DC fan

The Qt app serves as:

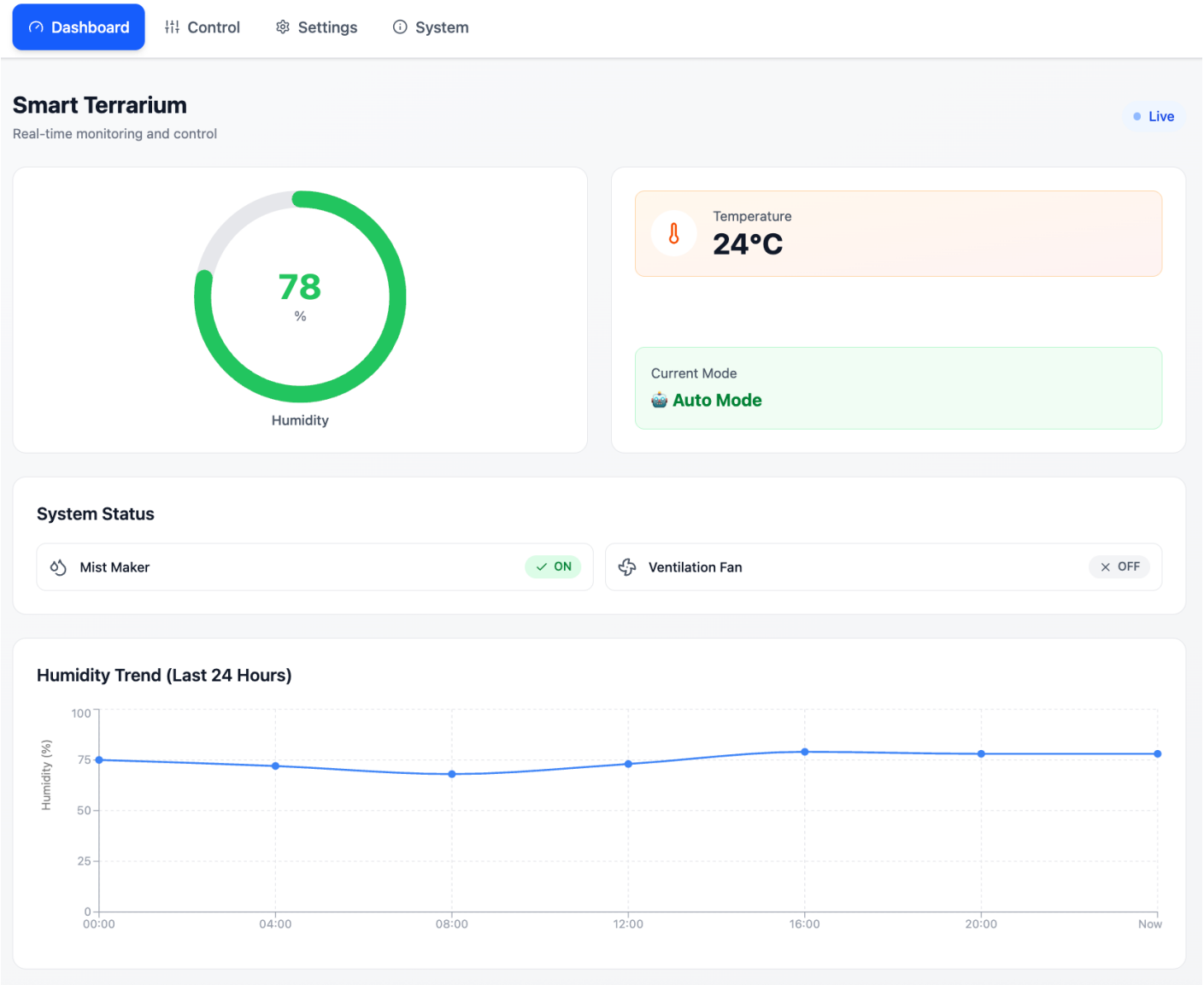
- Run Qt UI
- Implement business logic (thresholds, automatic control)
- Read sensor data (via I²C/SPI/UART)
- Control actuators (GPIO)
- Data logging (internal flash or SD card)
- Communication with optional Wi-Fi module

1.2 Screen Layout:

- **Main Dashboard Screen**
  - Circular humidity gauge
  - Status indicators: Mist Maker ON/OFF, Fan ON/OFF, Auto/Manual Mode
  - Real-time humidity trend graph
  - Current temperature (if sensor supports it)
- **Control Screen**
  - Manual ON/OFF toggle buttons
  - Auto mode enable/disable
  - "Boost Humidity" button (temporary override)
- **Settings Screen**
  - Humidity target range (e.g., 70–85%)
  - Cycle timing parameters
  - Sensor calibration
  - Display brightness

  - Wi-Fi (if supported)

- **System Screen**

  - Device status
  - Logs (flash or SD)
  - Firmware version
  - Factory reset

2 Wireframes